Gym Membership Analytics Portfolio Project

By: Larson St-Cyr

Project Links

- Tableau Public Dashboard: https://public.tableau.com/app/profile/larson.st.cyr/viz/GymMembershipAnalytics/GymMembershipAnalyticsDashboard
- GitHub Repository: https://github.com/lstcyr25/Gym-Membership-Analytics/tree/main

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Introduction

This project explores gym membership and member activity data to uncover trends related to member engagement, gym utilization, and service usage through data analytics and business intelligence techniques.

Working in a gym environment sparked my interest in analyzing operational and behavioral patterns using data rather than assumptions. The objective of this project was to simulate a real-world analytics workflow by performing data cleaning, validation, exploratory analysis, and dashboard development using industry-standard tools.

The dataset was sourced from Kaggle and transformed into an interactive Tableau dashboard designed to communicate actionable business insights.

Project Objectives

The primary goals of this project were to:

- Perform data cleaning and validation using SQL Server
- Analyze gym member behavior and operational trends
- Build interactive dashboards in Tableau Public
- Demonstrate an end-to-end ETL workflow

Tools & Technologies Used

- Microsoft Excel - Initial staging and pre-cleaning
- SQL Server Management Studio (SSMS) - Data cleaning, validation, and exploratory analysis
- Tableau Public - Dashboard development and visualization
- GitHub - Documentation and version control

Dataset Overview

The dataset contains information related to gym members, attendance behavior, service usage, and demographics.

Dataset Features Included

- Membership types
- Weekly attendance frequency
- Check-in and check-out times
- Average time spent in the gym
- Group fitness participation
- Personal training usage
- Sauna usage
- Beverage purchases
- Demographic information

Data Preparation & Cleaning Process

The project began with an initial review of the dataset in Microsoft Excel. During this stage, duplicate records were checked, column formatting was reviewed, and the dataset was prepared for SQL import.

After importing the data into SQL Server Management Studio, a staging table was created to preserve the original dataset before performing any transformations or validation processes.

Data Cleaning Steps Performed

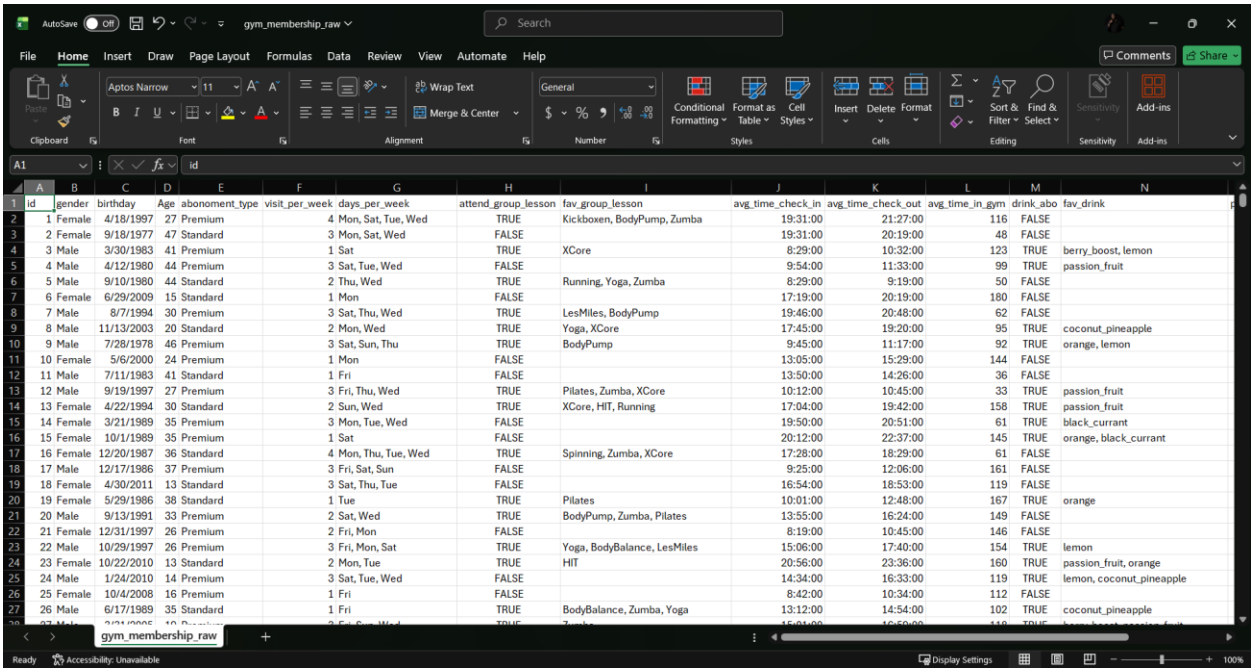
- Checked for NULL and missing values
- Standardized categorical fields
- Validated time-based records
- Verified attendance consistency
- Identified invalid weekly visit values
- Performed service usage validation checks
- Prepared fields for Tableau visualization

Example Validation Checks

- Members with weekly visits greater than 7
- Check-out times occurring before check-in times

- Members assigned trainers without personal training enabled

This process ensured the dataset was accurate, structured, and suitable for analysis.



id	gender	birthday	Age	abonoment_type	visit_per_week	days_per_week	attend_group_lesson	fav_group_lesson	avg_time_check_in	avg_time_check_out	avg_time_in_gym	drink_abo	fav_drink
1	Female	4/18/1997	27	Premium	4	Mon, Sat, Tue, Wed	TRUE	Kickboxen, BodyPump, Zumba	19:31:00	21:27:00	116	FALSE	
2	Female	9/18/1977	47	Standard	3	Mon, Sat, Wed	FALSE		19:31:00	20:19:00	48	FALSE	
3	Male	3/30/1983	41	Premium	1	Sat	TRUE	XCore	8:29:00	10:32:00	123	TRUE	berry_boost, lemon
4	Male	4/12/1980	44	Premium	3	Sat, Tue, Wed	FALSE		9:54:00	11:33:00	99	TRUE	passion_fruit
5	Male	9/10/1980	44	Standard	2	Thu, Wed	TRUE	Running, Yoga, Zumba	8:29:00	9:19:00	50	FALSE	
6	Female	6/29/2009	15	Standard	1	Mon	FALSE		17:19:00	20:19:00	180	FALSE	
7	Male	8/7/1994	30	Premium	3	Sat, Thu, Wed	TRUE	LesMiles, BodyPump	19:46:00	20:48:00	62	FALSE	
8	Male	11/13/2003	20	Standard	2	Mon, Wed	TRUE	Yoga, XCore	17:45:00	19:20:00	95	TRUE	coconut_pineapple
9	Male	7/28/1978	46	Premium	3	Sat, Sun, Thu	TRUE	BodyPump	9:45:00	11:17:00	92	TRUE	orange, lemon
10	Female	5/6/2000	24	Premium	1	Mon	FALSE		13:05:00	15:29:00	144	FALSE	
11	Male	7/11/1983	41	Standard	1	Fri	FALSE		13:50:00	14:26:00	36	FALSE	
12	Male	9/19/1997	27	Premium	3	Fri, Thu, Wed	TRUE	Pilates, Zumba, XCore	10:12:00	10:45:00	33	TRUE	passion_fruit
13	Female	4/22/1994	30	Standard	2	Sun, Wed	TRUE	XCore, HIT, Running	17:04:00	19:42:00	158	TRUE	passion_fruit
14	Female	3/21/1989	35	Premium	3	Mon, Tue, Wed	FALSE		19:50:00	20:51:00	61	TRUE	black_currant
15	Female	10/1/1989	35	Premium	1	Sat	FALSE		20:12:00	22:37:00	145	TRUE	orange, black_currant
16	Female	12/20/1987	36	Standard	4	Mon, Thu, Tue, Wed	TRUE	Spinning, Zumba, XCore	17:28:00	18:29:00	61	FALSE	
17	Male	12/17/1986	37	Premium	3	Fri, Sat, Sun	FALSE		9:25:00	12:06:00	161	FALSE	
18	Female	4/30/2011	13	Standard	3	Sat, Thu, Tue	FALSE		16:54:00	18:53:00	119	FALSE	
19	Female	5/29/1986	38	Standard	1	Tue	TRUE	Pilates	10:01:00	12:48:00	167	TRUE	orange
20	Male	9/13/1991	33	Premium	2	Sat, Wed	TRUE	BodyPump, Zumba, Pilates	13:55:00	16:24:00	149	FALSE	
21	Female	12/31/1997	26	Premium	2	Fri, Mon	FALSE		8:19:00	10:45:00	146	FALSE	
22	Male	10/29/1997	26	Premium	3	Fri, Mon, Sat	TRUE	Yoga, BodyBalance, LesMiles	15:06:00	17:40:00	154	TRUE	lemon
23	Female	10/22/2010	13	Standard	2	Mon, Tue	TRUE	HIT	20:56:00	23:36:00	160	TRUE	passion_fruit, orange
24	Male	1/24/2010	14	Premium	3	Sat, Tue, Wed	FALSE		14:34:00	16:33:00	119	TRUE	lemon, coconut_pineapple
25	Female	10/4/2008	16	Premium	1	Fri	FALSE		8:42:00	10:34:00	112	FALSE	
26	Male	6/17/1989	35	Standard	1	Fri	TRUE	BodyBalance, Zumba, Yoga	13:12:00	14:54:00	102	TRUE	coconut_pineapple

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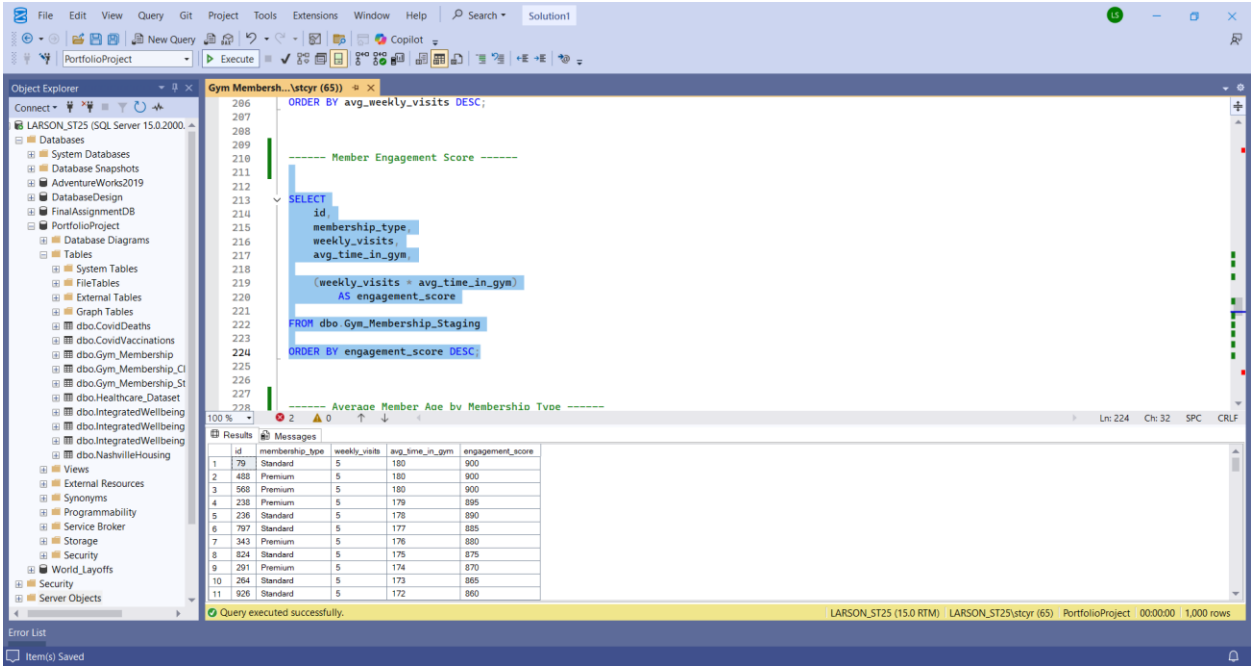
Exploratory Data Analysis

Exploratory analysis was performed using SQL queries to identify trends in member behavior and gym operations.

Areas of Analysis

- Membership distribution
- Weekly visit trends by age group
- Peak gym utilization hours
- Personal training adoption
- Sauna and beverage service usage
- Group fitness participation
- Member engagement behavior
- Trainer demand trends

The SQL analysis allowed patterns and relationships within the dataset to be identified before visualization.



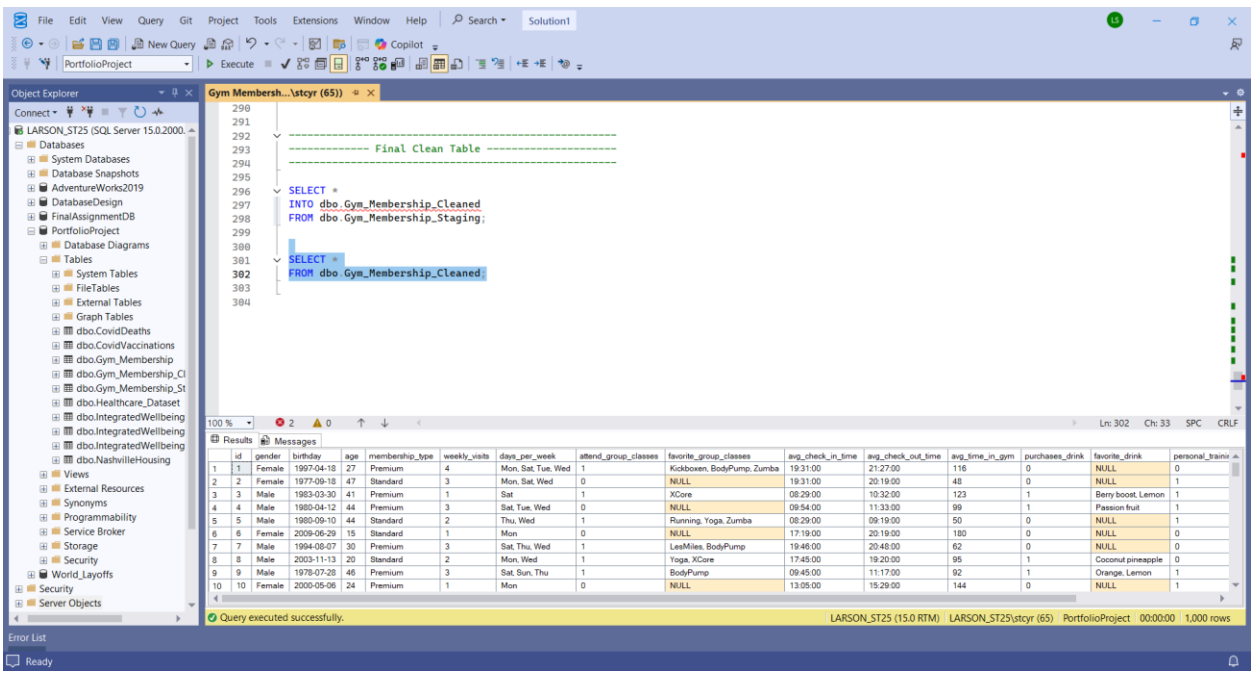


Tableau Dashboard Development

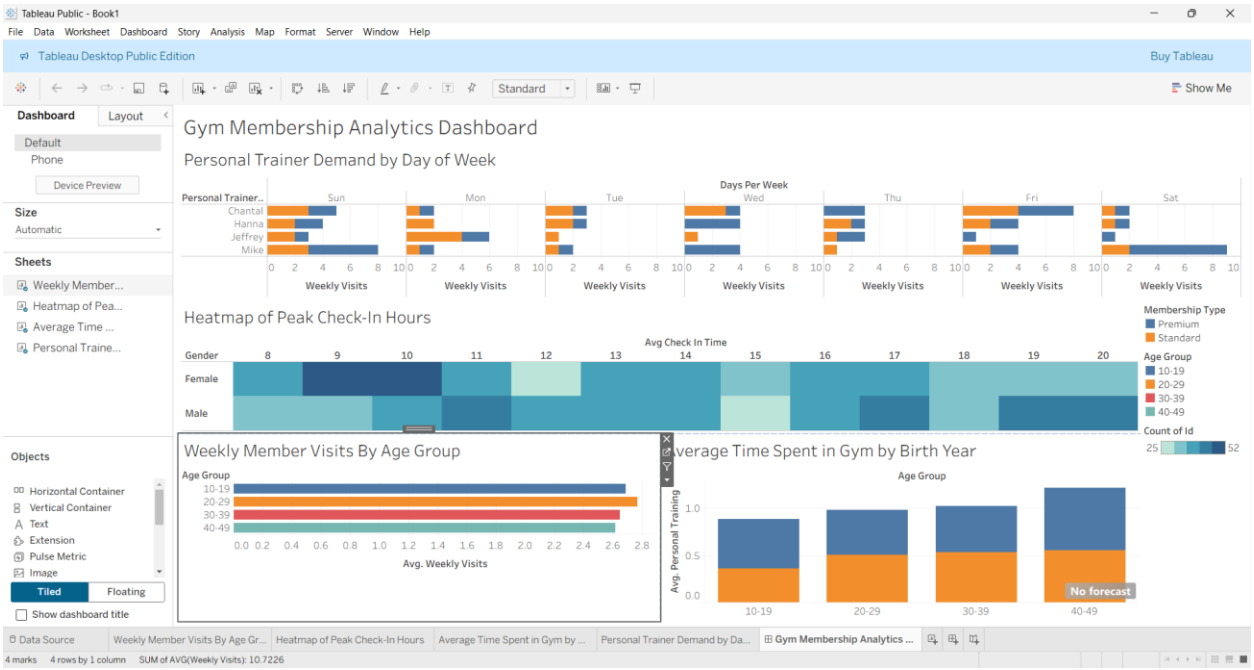
An interactive Tableau dashboard was created to visually communicate operational and behavioral insights from the dataset.

Dashboard Features

- Membership distribution analysis

- Weekly visits by demographic groups
- Peak gym utilization heatmaps
- Service usage comparisons
- Personal training trends
- Group fitness participation analysis
- Operational behavior insights

Interactive filters were also implemented to improve user experience and allow dynamic analysis across multiple categories.



Key Insights

Several meaningful trends were identified throughout the analysis process:

- Members between the ages of 20–29 demonstrated the highest average weekly gym attendance, while older age groups showed slightly lower visit frequency.
- Peak check-in activity occurred during morning and evening hours, indicating the busiest operational periods for gym utilization.
- Personal trainer demand varied throughout the week, with certain trainers consistently maintaining higher member engagement levels.

These findings demonstrate how data analytics can support operational decision-making and improve understanding of member behavior.

What I Learned

This project strengthened my understanding of the complete analytics workflow from raw data to business insight.

Through this project, I gained practical experience with:

- SQL data cleaning and transformation
- Exploratory data analysis
- Data validation techniques
- Tableau dashboard design
- Data storytelling and visualization
- Structuring analytics projects for business audiences
- Applying analytical thinking to operational questions

Additionally, working with a gym-focused dataset allowed me to connect real-world experience with technical analytics skills.

Conclusion

This project demonstrates an end-to-end business intelligence workflow using Excel, SQL Server, and Tableau Public. By combining data cleaning, exploratory analysis, and visualization techniques, the project transformed raw gym membership data into meaningful operational insights.

The experience strengthened both my technical analytics abilities and my understanding of how data can be used to support business decision-making in real-world environments.